

Business Finance (AF2004)

Assignment 1

1. If you deposit \$2,000 in a bank account that pays 6% interest annually, how much will be in your account after 5 years?
2. What is the present value of a security that will pay \$29,000 in 20 years if securities of equal risk pay 5% annually?
3. If you deposit money today in an account that pays 4% annual interest, how long will it take to double your money?
4. Your parents will retire in 19 years. They currently have \$350,000 saved, and they think they will need \$800,000 at retirement. What annual interest rate must they earn to reach their goal, assuming they don't save any additional funds?
5. Find the following values. Compounding/discounting occurs annually.
 - a. An initial \$200 compounded for 10 years at 4%
 - b. An initial \$200 compounded for 10 years at 8%
 - c. The present value of \$200 due in 10 years at 4%
 - d. The present value of \$1,870 due in 10 years at 8% and at 4%
 - e. Define present value and illustrate it using a time line with data from part d. How are present value affected by interest rate?
6. Find the following values using the equations. Compounding/discounting occurs annually.
 - a. An initial \$600 compounded for 1 year at 6%
 - b. An initial \$600 compounded for 2 years at 6%
 - c. The present value of \$600 due in 1 year at a discount rate of 6%
 - d. The present value of \$600 due in 2 years at a discount rate of 6%